

# Project Design Phase-II

## Technology Stack (Architecture & Stack)

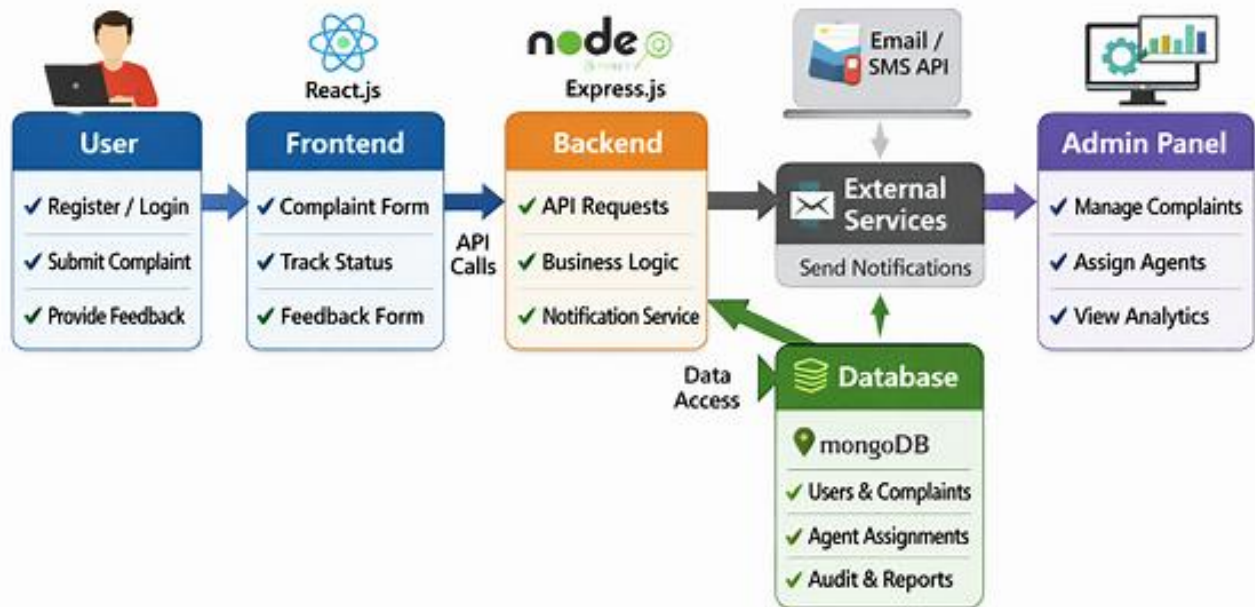
|               |                           |
|---------------|---------------------------|
| Date          | 18 June 2026              |
| Team ID       |                           |
| Project Name  | Online Complaint Register |
| Maximum Marks | 4 Marks                   |

## Technical Architecture – Online Complaint Register (MERN)

### Diagram (Conceptual Flow)

- **Frontend (React.js)** → User submits complaints, registers/logs in, views dashboard.
- **Backend (Node.js + Express.js)** → Handles authentication, complaint routing, notifications, and admin logic.
- **Database (MongoDB)** → Stores users, complaints, agents, feedback, and analytics data.
- **External Services** → Email/SMS APIs for notifications, cloud hosting for scalability.
- **Admin Panel** → Provides monitoring, reporting, and analytics.

## Online Complaint Register – Workflow Diagram



The **step-by-step flow** of how the system operates:

- **User side:** Register/Login → Submit Complaint → Provide Feedback.
- **Frontend (React.js):** Complaint form, status tracking, feedback form.
- **Backend (Node.js + Express.js):** API requests, business logic, notifications.
- **Database (MongoDB):** Stores users, complaints, agents, feedback, and reports.
- **External Services:** Email/SMS APIs for notifications.
- **Admin Panel:** Complaint management, agent assignment, analytics.



Table-1: Functional Modules

| Module                    | Description                                                              |
|---------------------------|--------------------------------------------------------------------------|
| User Registration & Login | Supports form, Gmail, LinkedIn, and Facebook authentication.             |
| Complaint Submission      | Users can submit complaints with category, description, and attachments. |
| Complaint Tracking        | Real-time status updates visible on dashboard.                           |
| User Confirmation         | Email/OTP confirmation after registration.                               |
| Admin Dashboard           | Assign complaints, monitor resolution, generate reports.                 |
| Notifications             | Automated alerts via email/SMS for complaint updates.                    |
| Feedback Module           | Users provide ratings and comments after resolution.                     |
| Analytics & Reports       | Insights into complaint trends, resolution times, and agent performance. |



Table-2: Non-Functional Attributes

| Attribute    | Description                                               |
|--------------|-----------------------------------------------------------|
| Usability    | Responsive UI, intuitive navigation, accessible design.   |
| Security     | JWT authentication, encrypted storage, role-based access. |
| Reliability  | Error handling, backup mechanisms, consistent uptime.     |
| Performance  | Optimized APIs and queries, <2s response time.            |
| Availability | 99% uptime with cloud deployment and load balancing.      |
| Scalability  | Supports growth across multiple departments/cities.       |

# Application Characteristics

| Characteristic           | Description                                                                       |
|--------------------------|-----------------------------------------------------------------------------------|
| Open-Source Frameworks   | MERN stack – MongoDB, Express.js, React.js, Node.js.                              |
| Security Implementations | JWT authentication, bcrypt password hashing, CORS, Helmet, OWASP guidelines.      |
| Scalable Architecture    | Three-tier architecture with microservices support for future expansion.          |
| Availability             | High availability through cloud deployment and load balancing.                    |
| Performance              | Optimized API calls, MongoDB indexing, React lazy loading, CDN for static assets. |